

Machine Learning Introduction 2

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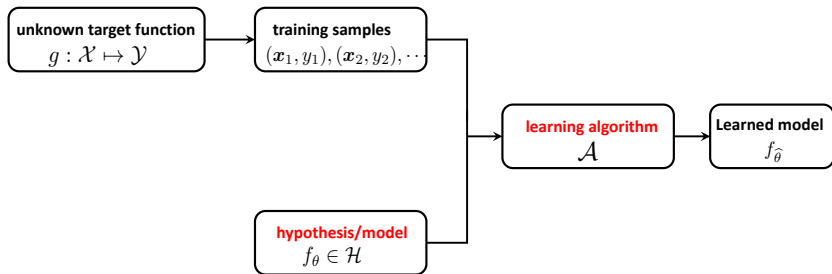


Recap: Supervised Learning Components

Components:

- ▶ Target function $g : \mathcal{X} \rightarrow \mathcal{Y}$ (underlying credit approval model)
- ▶ Training dataset: $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$ (historical records)
- ▶ Hypothesis space $\mathcal{H}$ (learning scope to approximate g)
- ▶ Hypothesis/model: f_{θ} (model to be determined)
- ▶ Optimization algorithm: $\mathcal{A}$ (learning the model from data)

Recap: Supervised Learning: High-level View



Hopefully,

$$f_{\hat{\theta}} \approx g$$

Predict/decision: When a new sample data (test data) x comes, the label is predicted as:

$$y \leftarrow f_{\hat{\theta}}(x).$$

$\rightsquigarrow$ We now discuss linear models for classification and regression.

Agenda

Linear Classification I — The Perceptron

Linear Regression — Least Squares

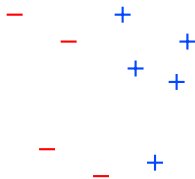
Maximum Likelihood Estimation (MLE)

MLE Interpretation of Least Squares

Feasibility of Learning

Classification

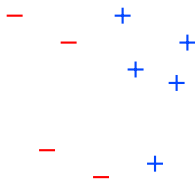
Consider the following illustration of the credit approval example:



- In the above figure, we can regard each point as one applicant's data (sample). This forms the training data pairs $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$ with $y_i = \pm 1$ (approve or deny).

Classification

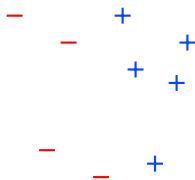
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- ▶ **Classification** is to learn a model $f_{\theta} : \mathbb{R}^d \rightarrow \{-1, +1\}$ based on the training samples to correctly classify the training samples (to approximate the underlying unknown g). This is called **binary classification** since we only have two classes.

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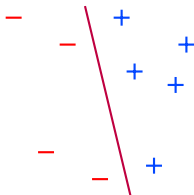


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What type of model f_{θ} can we choose?

Linear Classification

Consider the following illustration of the credit approval example:



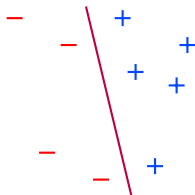
Linear classification: Find a linear model based on the training sample to correctly classify the training samples. Namely, f_{θ} takes the linear form

$$f_{\theta}(\mathbf{x}) = \boldsymbol{\theta}^{\top} \mathbf{x} + b.$$

Correspondingly, the hypothesis space is given by

$$\mathcal{H} := \{f_{\theta}(\mathbf{x}) = \boldsymbol{\theta}^{\top} \mathbf{x} + b : \boldsymbol{\theta} \in \mathbb{R}^n, b \in \mathbb{R}\},$$

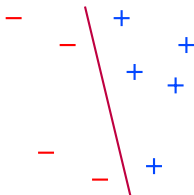
Linear Classification



The linear model is learned to separate the dataset:

$$\begin{cases} f_{\theta}(\mathbf{x}_i) > 0 & \text{if } y_i = +1, \\ f_{\theta}(\mathbf{x}_i) < 0 & \text{if } y_i = -1, \end{cases} \quad \forall i = 1, \dots, n.$$

Linear Classification



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The classification result can be represented using the “sign” function, i.e.,

$$y_i = \text{sign}(f_{\theta}(\mathbf{x}_i)),$$

where

$$\text{sign}(z) = \begin{cases} +1, & \text{if } z > 0, \\ -1, & \text{if } z < 0, \\ [-1, +1], & \text{if } z = 0. \end{cases}$$

Absorbing the Bias Term

- ▶ The term “ b ” in the linear model $f_{\boldsymbol{\theta}}(\mathbf{x}) = \boldsymbol{\theta}^{\top} \mathbf{x} + b$ is called **bias** term.
- ▶ Let $\theta_0 = b$ and $x_0 = 1$, and now, we define

$$\tilde{\mathbf{x}} = (x_0, \mathbf{x}) \in \mathbb{R}^{d+1}$$

$$\tilde{\boldsymbol{\theta}} = (\theta_0, \boldsymbol{\theta}) \in \mathbb{R}^{d+1}$$

It then follows that

$$\tilde{\boldsymbol{\theta}}^{\top} \tilde{\mathbf{x}} = \boldsymbol{\theta}^{\top} \mathbf{x} + \theta_0 x_0 = \boldsymbol{\theta}^{\top} \mathbf{x} + b.$$

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In the sequel, we will use $\boldsymbol{\theta}$ and $\mathbf{x}$ to represent $\tilde{\boldsymbol{\theta}}$ and $\tilde{\mathbf{x}}$ to ease notation, respectively. Then, we can compactly write the linear model as

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \boldsymbol{\theta}^{\top} \mathbf{x} \quad \text{and} \quad y = \text{sign}(f_{\boldsymbol{\theta}}(\mathbf{x})).$$

In this way, the bias term is absorbed and the linear model representation is more compact.

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↪ After choosing the model, we need an algorithm to find the classifier.

Linear Classification Algorithm: The Perceptron

- ▶ The **perceptron** was invented in 1943 by McCulloch and Pitts.
- ▶ It is a simple linear classification algorithm used to find a separation line based on the training dataset.
- ▶ The perceptron was intended to be a machine, rather than a program, and while its first implementation was in software for the IBM 704.
- ▶ Later in 1957, it was implemented in hardware-level machine and the machine is called "Mark I perceptron", designed for image recognition.

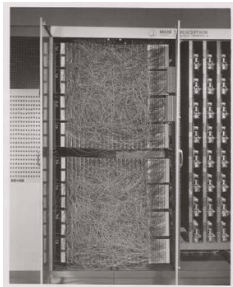


Figure: Mark I Perceptron machine.

The Perceptron: Algorithm Procedure

- ▶ Given the training dataset $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$, with $y_i = \pm 1$.
- ▶ Linear classification model

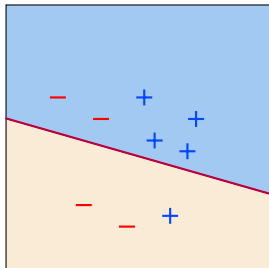
$$f_{\theta}(\mathbf{x}) = \theta^{\top} \mathbf{x}, \quad y = \text{sign}(f_{\theta}(\mathbf{x})).$$

- ▶ **Intuition:** θ should be in the same direction as the points $\mathbf{x}_i$ in the $y = +1$ class, and in the opposite direction as those in the $y = -1$ class
- ▶ **Optimization:** Pick a **misclassified** data $\mathbf{x}_i$ (any misclassified one)

$$\underbrace{\text{sign}(f_{\theta}(\mathbf{x}_i))}_{=\text{sign}(\theta^{\top} \mathbf{x}_i)} \neq y_i.$$

- ▶ Update rule

$$\theta = \theta + y_i \mathbf{x}_i.$$

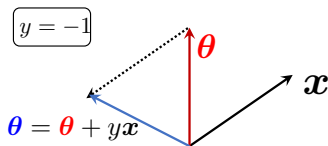
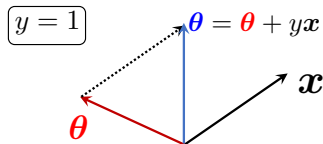


The Perceptron: Interpretation

► Update rule

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- This update rule pushes the line to correctly classify the picked data. Iteratively repeat this procedure, until all training samples are correctly classified.

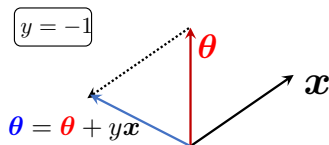
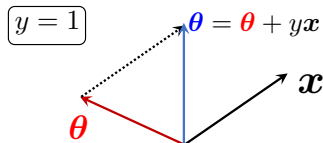


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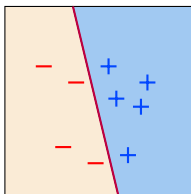
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- This update rule pushes the line to correctly classify the picked data. **Iteratively** repeat this procedure, until all training samples are correctly classified.



Learned model:



Example

Which statement is true?

- (A) The perceptron algorithm is used for unsupervised classification.
- (B) The perceptron algorithm can find a classifier (separation line) in one update step.
- (C) The perceptron algorithm will stop at the same separation line for every possible runs.
- (D) The perceptron is tailored for binary classification.

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Answer: (D). For (A), we have label, hence it is supervised. Actually, classification always has label and is always supervised. For (B), it depends on the data. It may take many iterative steps to converge. For (C), it may stop at different separation lines, depending on which misclassified data to choose.

Discussions

From the perceptron learning algorithm, we still have the following questions:

- ▶ The learned model successfully classifies the training data points. Does this mean that it will perform well for the **future/test** data points?
- ▶ Is the learned model $f_{\hat{\theta}}$ **optimal** in terms of classification?
- ▶ Is perceptron guaranteed to converge to a separation line?
- ▶ How about the training samples are **not linearly separable**? Namely, the two classes have an intersection. Does perceptron still work?

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The first question will be answered in the training versus testing section.

The second question will be answered in the section of support vector machine.

The third and forth questions will be answered in our Homework 1.

↪ We will have tutorial 2 (Sep 16) to introduce how to implement perceptron, which will be helpful for completing Homework 1.

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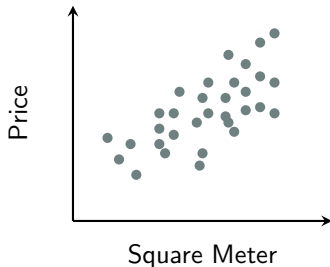
Feasibility of Learning

Regression

Data: Given past sales data $\{(x_i, y_i)\}_{i=1}^n$, where

$x_i \in \mathbb{R}$ represents square meter

y_i represents apartment price



Regression is to learn a model f_θ that captures relationship between x_i and y_i . Given a new apartment with square meter x , predict $y = f_\theta(x)$.

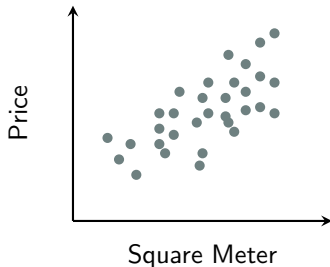
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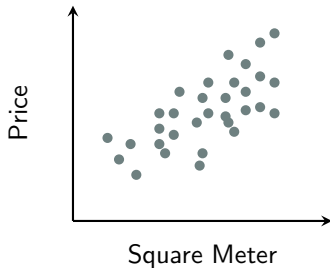
What is the difference between regression and classification?

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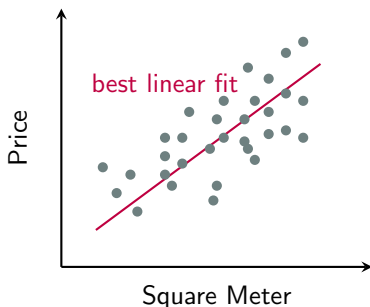
Regression means that y is real-valued (continuous).

What is the difference between regression and classification? In regression, y is continuous, while in classification it is categorical.

Linear Regression

- ▶ If we choose the hypothesis space to be linear model, then we have **linear regression**. That is, we assume that the apartment price is a **linear** function of square meter.
- ▶ For the previous simple example, the linear model can be expressed as

$$y_i \approx f_{\theta}(x_i) = \theta x_i.$$



Multi-dimensional Linear Regression

Data: Given past sales data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where

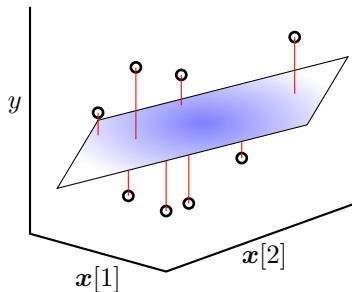
$\mathbf{x}_i \in \mathbb{R}^d$ represents square meter, metro, schools...

y_i represents price

Linear model:

$$y_i \approx f_{\boldsymbol{\theta}}(\mathbf{x}_i) = \boldsymbol{\theta}^{\top} \mathbf{x}_i,$$

where $\boldsymbol{\theta} \in \mathbb{R}^d$.



How to find the best linear fit? Least squares.

Least Squares

The learning problem can be formulated as:

$$\hat{\theta} = \operatorname{argmin}_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(\mathbf{x}_i), y_i)$$

Since our goal is to enforce $y_i \approx f_{\theta}(\mathbf{x}_i)$ and the label y is continuous real-valued. Thus, we can choose the most common ℓ_2 -loss function:

$$\ell(a, b) = (a - b)^2.$$

The choice of ℓ_2 -loss results in the famous least squares (LS) formulation:

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↪ We can also select absolute value loss, leading to robust linear regression formulation (Homework 1).

Least Squares in Matrix Notation

Matrix representation of data and label:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n$$

Fact:

$$\begin{bmatrix} \boldsymbol{\theta}^\top \mathbf{x}_1 - y_1 \\ \vdots \\ \boldsymbol{\theta}^\top \mathbf{x}_n - y_n \end{bmatrix} = \mathbf{X}\boldsymbol{\theta} - \mathbf{y}$$

Least squares:

$$\min_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n (\boldsymbol{\theta}^\top \mathbf{x}_i - y_i)^2 \iff \boxed{\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2}$$

▪

Solution of Least Squares

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Let (we can ignore the $\frac{1}{n}$ without loss of generality)

$$\mathcal{L}(\boldsymbol{\theta}) = \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Then

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) &= (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) \\ &= \boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}\end{aligned}$$

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Facts:

- ▶ If $h(\boldsymbol{\theta}) = \mathbf{c}^\top \boldsymbol{\theta}$, then $\nabla h(\boldsymbol{\theta}) = \mathbf{c}$.
- ▶ If $h(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{M} \boldsymbol{\theta}$ ($\mathbf{M}$ is symmetric), then: $\nabla h(\boldsymbol{\theta}) = 2\mathbf{M}\boldsymbol{\theta}$.

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Take the gradient

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = 2\mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

Solution of Least Squares

Let (we can ignore the $\frac{1}{n}$ without loss of generality)

$$\mathcal{L}(\theta) = \|X\theta - y\|_2^2$$

Then

$$\begin{aligned}\mathcal{L}(\theta) &= (X\theta - y)^\top (X\theta - y) \\ &= \theta^\top X^\top X\theta - 2\theta^\top X^\top y + y^\top y\end{aligned}$$

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- ▶ If $h(\theta) = c^\top \theta$, then $\nabla h(\theta) = c$.
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Take the gradient

$$\nabla_{\theta} \mathcal{L}(\theta) = 2X^\top (X\theta - y)$$

Setting the gradient to zero and due to **convexity** of $\mathcal{L}$, the optimal solution $\hat{\theta}$ satisfies

$$X^\top X\hat{\theta} = X^\top y$$

How to solve for $\hat{\theta}$?

Solution of Least Squares: Case I

Case I: $\mathbf{X} \in \mathbb{R}^{n \times d}$ has full column rank (its rank is d)

- ▶ $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}$ will be full rank / invertible (Homework 1).
- ▶ We have

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{y}.$$

Pseudo-inverse of matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ when it has full column rank is defined as

$$\mathbf{X}^\dagger = \left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \in \mathbb{R}^{d \times n}.$$

- ▶ We have $\mathbf{X}^\dagger \mathbf{X} = \mathbf{I}$, but not necessarily $\mathbf{X} \mathbf{X}^\dagger = \mathbf{I}$.

Hence, we can also write

$$\hat{\boldsymbol{\theta}} = \mathbf{X}^\dagger \mathbf{y}.$$

Solution of Least Squares: Case I

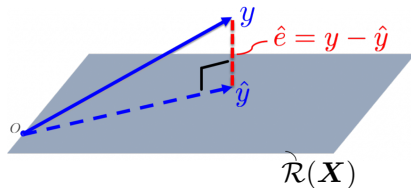
Geometric interpretation:

- Let

$$\mathcal{P}_X = \mathbf{X}\mathbf{X}^\dagger.$$

- $\mathcal{P}_X$ is the **projection matrix** onto the range space $\mathcal{R}(\mathbf{X})$: the space of all possible predictions $\mathbf{X}\boldsymbol{\theta}$ our linear model can make.
- What does the LS solution mean?

$$\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{y} - \mathbf{X}\mathbf{X}^\dagger\mathbf{y} = \mathbf{y} - \mathcal{P}_X(\mathbf{y}) := \mathbf{y} - \hat{\mathbf{y}}.$$



Conclusion: The LS solution is such that $\hat{\mathbf{y}} := \mathbf{X}\hat{\boldsymbol{\theta}} = \mathcal{P}_X(\mathbf{y})$ is the **orthogonal projection** of $\mathbf{y}$ onto the range space $\mathcal{R}(\mathbf{X})$.

Solution of Least Squares: Case II

Case II: $\mathbf{X} \in \mathbb{R}^{n \times d}$ does not have full column rank (its rank is $< d$).

- One typical case is, $n < d$.

Solution of Least Squares: Case II

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- ▶ One typical case is, $n < d$. **It means overfitting.**
- ▶ We do not have a unique solution. Instead, we have **infinitely many** solutions.

We can represent all the solutions using singular value decomposition (SVD), and it will be discussed in Homework 1.

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How to obtain a unique and meaningful solution in this case?

↪ Regularization, which we will study in details later in the overfitting chapter (simple derivation will be discussed in Homework 1).

↪ Next lecture: Maximum likelihood estimation and feasibility of learning.

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Maximum Likelihood Estimation: A Simple Example

Flip Coin

- ▶ Setup: For a 'special' coin (with head and back), flip it.
- ▶ Outcome: Dataset $\mathcal{D} = \{H, H, B, H, B, \dots\}$, k heads out of n flips.
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What is the probability it will be head?

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- ▶ The outcome of every flip follows a Bernoulli distribution:
 $\Pr[\text{head}] = \theta$ and $\Pr[\text{back}] = 1 - \theta$.
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 - The $i + 1$ -th flip is independent of i -th flip.
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Task: Learning θ from data using Maximum Likelihood Estimation (MLE).

Conditional Probability and Likelihood

- ▶ The conditional probability

$$\Pr[y|x]$$

means **the probability of y given x** . An important quantity for learning ($\rightsquigarrow$ later course on logistic regression and language modeling)

- ▶ How about

$$\Pr[\mathcal{D}|\theta]?$$

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$$\Pr[\mathcal{D}|\theta]?$$

Likelihood of data

$\Pr[\mathcal{D}|\theta]$: The probability of observed data given parameter θ .

Write down the likelihood:

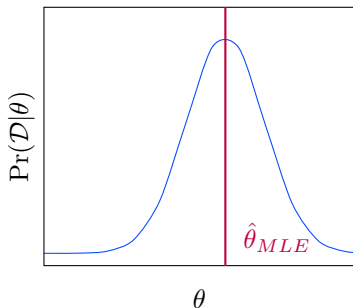
$$\Pr[\mathcal{D}|\theta] = \theta^k (1 - \theta)^{n-k}$$

since we have k heads out of n trials and i.i.d.

Maximum Likelihood Estimation

Maximum likelihood estimation: The principle

Maximize the probability of observed data over parameter θ .



Log-likelihood:

$$\log(\Pr[\mathcal{D}|\theta]) = k \log(\theta) + (n - k) \log(1 - \theta)$$

Maximum Likelihood Estimation

Set $\mathcal{L}(\theta) = \log(\Pr[\mathcal{D}|\theta])$, we have

$$\frac{\partial \mathcal{L}(\theta)}{\partial \theta} = \frac{k}{\theta} - \frac{n-k}{1-\theta}$$

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Why log-likelihood?

► $\log(\cdot)$ is increasing, thus

$$\max_{\theta} \Pr[\mathcal{D}|\theta] \iff \max_{\theta} \log(\Pr[\mathcal{D}|\theta])$$

► $\log(\Pr[\mathcal{D}|\theta])$ is often easier to maximize than $\Pr[\mathcal{D}|\theta]$.

Maximum Likelihood Estimation: General Procedure

- ▶ Observe (i.i.d.) data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ following $p(\mathbf{x}|\boldsymbol{\theta}^*)$.
- ▶ Build the likelihood function $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^n p(\mathbf{x}_i|\boldsymbol{\theta})$ for some parameter $\boldsymbol{\theta}$.
- ▶ Log-likelihood $\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \log(p(\mathbf{x}_i|\boldsymbol{\theta}))$
- ▶ Maximum likelihood estimator: $\hat{\boldsymbol{\theta}}_{MLE} = \operatorname{argmax}_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$.

Key point: Know $p(\mathbf{x}|\boldsymbol{\theta})$.

Agenda

Linear Classification I — The Perceptron

Linear Regression — Least Squares

Maximum Likelihood Estimation (MLE)

MLE Interpretation of Least Squares

Feasibility of Learning

Recall: Multivariate Gaussian distribution

- **Gaussian distribution.** A random variable X is said to follow $\mathcal{N}(\mu, \sigma^2)$ (Gaussian distribution with mean μ and variance σ^2) if its probability density function (PDF) is given by

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

- **Multivariate Gaussian distribution.** We say the random vector $\mathbf{X} \in \mathbb{R}^d$ follows Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (assumed to be PD), if its PDF is given by

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

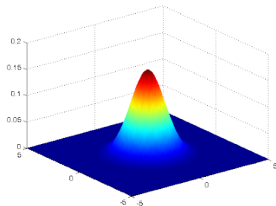
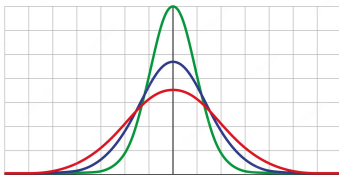
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LS for Linear Regression: MLE Interpretation

- ▶ Recap LS:

$$\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

- ▶ Recap our model:

$$y_i \approx f_{\boldsymbol{\theta}}(\mathbf{x}_i) = \boldsymbol{\theta}^\top \mathbf{x}_i.$$

- ▶ To be more explicit, consider

$$y_i = \boldsymbol{\theta}^\top \mathbf{x}_i + \epsilon_i \quad \text{assume} \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where ϵ_i are i.i.d. Gaussian for $i = 1, \dots, n$.

- ▶ In matrix notation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon} \quad \text{with} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Sigma} = \text{diag}(\sigma^2, \dots, \sigma^2)$.

LS for Linear Regression: MLE Interpretation

Equivalently,

$$\epsilon = \mathbf{y} - \mathbf{X}\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \Sigma).$$

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$$\boldsymbol{\epsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}).$$

The likelihood: The probability of observed data given parameters $\boldsymbol{\theta}$

$$\begin{aligned} p(\mathcal{D}|\boldsymbol{\theta}) &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}\boldsymbol{\epsilon}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\epsilon}\right) \quad \leftarrow \text{PDF of Gaussian distribution} \\ &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})\right). \end{aligned}$$

Log-likelihood:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= \log(p(\mathcal{D}|\boldsymbol{\theta})) = \text{constant} - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \\ &= \text{constant} - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2. \end{aligned}$$

LS for Linear Regression: MLE Interpretation

MLE principle:

$$\hat{\boldsymbol{\theta}}_{MLE} = \operatorname{argmax}_{\boldsymbol{\theta}} -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 = \operatorname{argmin}_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2,$$

which is exactly LS.

Take-home message

Least square is a maximum likelihood estimator under Gaussian noise assumption

How Good Is This Estimator? Unbiased or Biased.

Suppose the target hypothesis g is $g(\mathbf{X}) = \mathbf{X}\boldsymbol{\theta}^*$ and

$$\mathbf{y} = g(\mathbf{X}) + \epsilon = \mathbf{X}\boldsymbol{\theta}^* + \epsilon \quad \text{with} \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Goal: To learn the target $\boldsymbol{\theta}^*$ (then we know g to the most extent).

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Definition: An estimator of a given parameter is said to be **unbiased** if its expected value is equal to the true/underlying value of the parameter.

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Unbiased Estimator:

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_{MLE}] = \boldsymbol{\theta}^*$$

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Feasibility of Learning

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Is Learning Possible? When and How?

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We briefly answer the first question. So, the question is: Does training say anything about testing? (Think about the perceptron)

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- ▶ The (**really**) **unknown** target function g is the object of learning. Learning is all about to infer g **outside of the seen training dataset**.
- ▶ There are extreme cases, e.g., we know g on the training dataset, but nothing on other unseen (test) data points. Learning in this case is obviously infeasible.
- ▶ Know things we have already seen, this is not learning, it is **memorizing**.

Learning is Possible in a Probabilistic Way

From the previous example, we know that learning is not feasible if we

- ▶ make no assumptions about the connections between the training and test data points, and
- ▶ want to firmly (**deterministically**) predict something about the test data.

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From the previous example, we know that learning is not feasible if we

- ▶ make no assumptions about the connections between the training and test data points, and
- ▶ want to firmly (**deterministically**) predict something about the test data.

Fortunately, learning will be possible if we

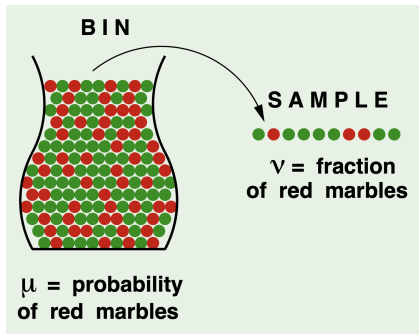
- ▶ make some assumptions that the training and testing data are related in some way. The most common assumption:

The training and test data are independent and identically distributed (i.i.d.)

- ▶ predict something about the test data **in a probabilistic way**.

A Bin Sampling Example: i.i.d. and Randomness

- ▶ Consider a bin with **red** and **green** marbles.
 - $\Pr[\text{red marble}] = \mu$.
 - $\Pr[\text{green marble}] = 1 - \mu$.
- ▶ The learning task is to learn μ .
- ▶ μ is **unknown** to us.
- ▶ We pick n marbles **randomly** and **independently** (with **replacement**). This means i.i.d.



The **fraction** of red marbles is v .

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It is “**Possible** versus **Probable**”.

- ▶ **No.** **Learning is not possible** if you want to be absolutely certain
 - ▶ There is some probability that the samples are mostly **green**, where learning fails.
- ▶ **Yes.** **Learning is possible** in a probabilistic way.
 - ▶ The sample frequency ν is close to μ with high probability.

Concentration Inequality

We can use tools from statistics to establish this “probable” learning result, i.e., **concentration inequality**. That is, to quantify “with high probability”.

We can have

$$\Pr[|\nu - \mu| > \varepsilon] \leq 2e^{-2n\varepsilon^2} \quad \forall \varepsilon > 0.$$

This result is immediately from **Hoeffding's inequality**.

- ▶ The result is valid for any n and ε .
- ▶ With proper n and ε , it says that ν is close to μ **with high probability** (i.e., probably). That is, $\nu \approx \mu$ is **probably and approximately correct (P.A.C.)**.

↪ In the sequel, we will study this training versus testing result for general machine learning such as classification and regression.